

Dr Yixuan Li

Research

Dr Yixuan Li is a researcher whose work bridges large language models (LLMs) and formal synthesis to develop user-friendly synthesis frameworks. His research has been published in top-tier venues spanning artificial intelligence (AI), natural language processing (NLP), programming languages, and verification.

Education

- 2022 – 2025  **PhD, Informatics (Computer Science)** (with full scholarship)
University of Edinburgh
Advised by [Elizabeth Polgreen](#) and [Michael O’Boyle](#)
Thesis: [Large Language Model Enabled Program Synthesis](#)
- 2020 – 2021  **MSc, Image and Video Communications and Signal Processing**
University of Bristol
Distinction (GPA: 76/100)
- 2014 – 2018  **BEng, Electronic Information**
Xidian University

Career

- Jan 2026 –  **Senior AI Engineer**, GSM Association (GSMA)
- Sep – Dec 2025  **Research Intern**, Huawei R&D UK, Edinburgh
Contributed to open-source projects including an AI browser, LLM-based code analysis, and code generation tools.
- Nov - Dec 2024  **Research Assistant**, Heriot-Watt University, Edinburgh
Explored LLM-assisted theorem proving within formal verification.
- Jan – Jun 2024  **Teaching Assistant**, System Design Project, University of Edinburgh
Supervised undergraduate teams building end-to-end robotic systems, from software design and embedded programming to physical prototyping and full robotic deployment.

Publications & Manuscripts

Accepted at PLDI, AAI, CAV, ACL (findings), FMCAD, IEEE TAC, SYNT

- 1 **Li, Y.**, Frampton, L., Mora, F., & Polgreen, E. (2025). Online Prompt Selection for Program Synthesis. *Proceedings of the AAI Conference on Artificial Intelligence*, 39(11), 11282–11289.
 [doi:10.1609/aaai.v39i11.33227](https://doi.org/10.1609/aaai.v39i11.33227)
- 2 **Li, Y.**, Magalhães, J. W. d. S., Brauckmann, A., O’Boyle, M. F. P., & Polgreen, E. (2025). Guided Tensor Lifting. *Proc. ACM Program. Lang.*, 9(PLDI), 1984–2006.  [doi:10.1145/3729330](https://doi.org/10.1145/3729330)
- 3 **Li, Y.**, Mora, F., Polgreen, E., & Seshia, S. A. (2023). Genetic Algorithms for Searching a Matrix of Metagrammars for Synthesis. *Workshop on Synthesis (SYNT)*.
 [doi:10.48550/arXiv.2306.00521](https://doi.org/10.48550/arXiv.2306.00521)

- 4 **Li, Y.**, Parsert, J., & Polgreen, E. (2024). Guiding Enumerative Program Synthesis with Large Language Models. In *Computer Aided Verification (CAV)* (pp. 280–301).
[doi:10.1007/978-3-031-65630-9_15](https://doi.org/10.1007/978-3-031-65630-9_15)
- 5 Tang, W., **Li, Y.**, Sypherd, C., Polgreen, E., & Belle, V. (2025, July). HyGenar: An LLM-Driven Hybrid Genetic Algorithm for Few-Shot Grammar Generation. In *Findings of the Association for Computational Linguistics: ACL 2025* (pp. 13640–13665).
[doi:10.18653/v1/2025.findings-acl.701](https://doi.org/10.18653/v1/2025.findings-acl.701)
- 6 Wang, Y., Zhu, Z., van Glabbeek, R., Zhang, J., & **Li, Y.** (2026). The similarity control problem with required events. *IEEE Transactions on Automatic Control*, 1–12.
[doi:10.1109/TAC.2026.3728353](https://doi.org/10.1109/TAC.2026.3728353)
- 7 Ye, L., **Li, Y.**, Frankel, G., Cheng, J., & Polgreen, E. (2025). Unlocking Hardware Verification with Oracle Guided Synthesis. In *Formal Methods in Computer-Aided Design (FMCAD)* (pp. 235–245). [doi:10.34727/2025/isbn.978-3-85448-084-6_30](https://doi.org/10.34727/2025/isbn.978-3-85448-084-6_30)

Service

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| Co-Organizer | ■ EuroProofNet Workshop on Theorem Proving in the Age of LLMs (2025) |
| Reviewer | ■ International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS, 2025) |
| | ■ EuroProofNet Workshop on Theorem Proving in the Age of LLMs (2025) |

Talks and Presentations

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| Dec 2025 | ■ European OpenHarmony Technical Forum, Edinburgh, UK |
| Sep 2025 | ■ Compilers Seminar, University of Edinburgh, UK |
| Jun 2025 | ■ PLDI Conference, Seoul, Korea |
| Feb 2025 | ■ AAI Conference, Philadelphia, USA |
| Nov 2024 | ■ Programming Languages Seminar, University of Bristol, UK |
| Oct 2024 | ■ Compilers Seminar, University of Edinburgh, UK |
| May 2024 | ■ LAIV AI Verification Seminar, Heriot-Watt University, UK |
| Mar 2024 | ■ EuroProofNet Workshop on Machine Learning in Proofs, Vienna, Austria |
| Jul 2023 | ■ SYNT Workshop, Paris, France |

Scholarships

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| 2022–2025 | ■ Fully Funded PhD Scholarship, The University of Edinburgh |
| May 2024 | ■ Verification Mentoring Workshop (VMW) Scholarship, CAV Conference |
| Mar 2016 | ■ University Scholarship, Xidian University |

Skills

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| Languages | ■ English, Mandarin, French, Japanese |
| Coding | ■ Python, MATLAB, C/C++, Java, JavaScript |